

ATUL PAHAL

Sonipat, Haryana

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TECHNICAL SKILLS

Languages: Python (Expert), JavaScript, HTML, CSS, C, Rust
ML & Data Science: TensorFlow, PyTorch, Scikit-learn, ONNX, Pandas, NumPy, Gradio, Matplotlib
Developer Tools: Neovim (LazyVim), Git/GitHub, Docker, Bun, VS Code, Linux, Hugging
Frameworks/Web: Langchain, FastAPI, ReactJS, NextJS, ExpressJS, Vite, Streamlit

CERTIFICATIONS

- Specialized Program in Artificial Intelligence & Machine Learning with Drone Tech, TiHAN IIT Hyderabad
- DCSP - HARTRON

PROJECTS

Oculus Vision: Real-time Object Detection | YOLO26, ONNX Runtime, Bun Jan 2026

- Architected a **privacy-first** vision app using **YOLO26 Nano**, performing real-time inference locally in the browser via **ONNX Runtime Web**.
- Implemented support for image, video, and live webcam analysis, achieving high-frame-rate detection without server-side processing.

AI-Based Crop Health Monitoring | Python, Random Forest, Scikit-Learn, Pandas Feb 2026

- Developed a Precision Agriculture model utilizing a **RandomForestClassifier** to monitor crop health with **92% accuracy** based on multispectral vegetation indices (NDVI, SAVI, EVI).
- Engineered a data pipeline to analyze spectral data and generate heatmaps for identifying "Healthy" vs. "Stressed" zones, enabling targeted fertilizer and water application.

Movie Recommender System | React, Vite, TMDB API, Similarity Scoring Jan 2026

- Developed a responsive recommendation engine utilizing **vector similarity scores** from a local dataset to suggest movies based on user selection.
- Integrated **TMDB API** for dynamic metadata fetching and poster rendering, optimized with **Vite** for fast production builds.

Breast Cancer Diagnostic System | Python, Scikit-learn, Gradio, Pandas Dec 2025

- Trained a **K-Nearest Neighbors (KNN)** classifier achieving **94.74% accuracy** on medical diagnostic data through feature engineering and standard scaling.
- Deployed the model via an interactive **Gradio** web interface, allowing real-time diagnosis prediction based on multidimensional medical features.

EDUCATION

Maharshi Dayanand University

B.A

Ongoing

Rohtak, India

DAV Centenary Pub Sch Samalkha Panipat Hry(CBSE)

Class 12

64%

Panipat, India

HOBBIES & INTERESTS

Linux Customization: Arch Linux, Hyprland, Dotfiles management, and Tiling Window Managers.
Automation: Developing custom Python and Bash scripts for workflow optimization.
Open Source: Exploring SOTA ML models on GitHub and customizing Neovim (LazyVim).